

1.4 Classwork: Standard Deviation — Markscheme

Calculator allowed. Marks are not printed on the student sheet; they are recorded here. Total: 18 marks.

Throughout: “standard deviation” means the population value σ_x , never the sample value s_x . The two sit side by side on the One-Variable Statistics screen, and Machine A in Problem 2 gives $s_x = 2$ exactly — which reads as more “intended” than the correct 1.87. This is the most likely error on the sheet.

1. [Maximum mark: 9]

$$n = \sum f = 50, \quad \sum fx = 175, \quad \sum fx^2 = 713.$$

GDC: menu $\rightarrow$ Statistics $\rightarrow$ Stat Calculations $\rightarrow$ One-Variable Statistics, with the x values in one list and the frequencies in a second list entered as the **Frequency List**.

$$(a) \bar{x} = \frac{\sum fx}{\sum f} = \frac{175}{50} = 3.5$$

M1
A1

M1 for a sum of fx divided by 50, by hand or by entering both lists on the GDC. A candidate who averages the six x values also reaches 3.5, by a method that fails as soon as the frequencies are not symmetric: award neither mark and say so in the return.

$$(b) \sigma_x = 1.4177 \dots = 1.42 \text{ (3 s.f.)}$$

A1

Accept 1.42, 1.418, or the full 1.4177... Do not accept $s_x = 1.4321 \dots$ (1.43), the sample standard deviation. Do not accept the variance 2.01 on its own.

$$(c) \text{Median} = 3, \quad Q_1 = 2, \quad Q_3 = 5$$

A1

$$\text{IQR} = Q_3 - Q_1 = 5 - 2 = 3$$

A1

All three values needed for the first A1; they are read directly from the One-Variable Statistics screen. FT: award the second A1 for $Q_3 - Q_1$ computed correctly from the candidate's own quartiles.

$$(d) \bar{x} + \sigma_x = 3.5 + 1.4177 \dots = 4.9177 \dots$$

M1

$$\text{Customers with } x = 5 \text{ or } x = 6: \quad 8 + 5 = 13 \text{ customers}$$

A1

M1 for forming $\bar{x} + \sigma_x$; A1 for 13. Accept the boundary quoted as 4.92. Counting $x \geq 5$ reaches the same 13 — accept it; counting both tails answers a different question — M1 only. FT: award the A1 for a count read correctly from the candidate's own $\bar{x} + \sigma_x$.

$$(e) \text{The mean increases, to } \frac{175 + 20}{51} = 3.82 \text{ (3 s.f.)}$$

A1

The standard deviation also increases, to 2.68 (3 s.f.), because 20 lies far above every other value and so is far from the mean; the spread of the data has grown. **R1**

Values are not required. A1 for stating that both increase; R1 for a reason referring to 20 lying far from the mean, or far from the rest of the data. “It is a big number” is not enough — distance from the mean is what drives the standard deviation, not size.

2. [Maximum mark: 9]

GDC: enter each machine's eight volumes as a list and run *One-Variable Statistics* on each.

(a) Machine A: $\bar{x} = 500$ **A1**

$\sigma_x = 1.8708 \dots = 1.87$ (3 s.f.) **A1**

Accept $\sqrt{3.5}$. **Do not accept 2** — that is s_x , the sample standard deviation. The two marks are independent: a candidate with the correct mean and $s_x = 2$ earns the first A1 only.

(b) Machine B: $\bar{x} = 500$ **A1**

$\sigma_x = 4.2720 \dots = 4.27$ (3 s.f.) **A1**

Accept $\sqrt{18.25}$. Do not accept $s_x = 4.57$.

(c) Machine A. **A1**

The two machines have the same mean, 500 ml, so the mean alone cannot separate them. Machine A's standard deviation is smaller (1.87 against 4.27), so its bottles are clustered more tightly around 500 ml. **R1**

R1 requires both statistics to be used: that the means are equal and that A's standard deviation is smaller. Naming Machine A with only "its standard deviation is smaller" earns the A1 and not the R1, because the question asked for both. FT: award both marks for a choice argued correctly from the candidate's own values in (a) and (b).

(d) New mean = $500 + 3 = 503$ ml **A1**

New standard deviation = 1.87 ml, unchanged. **A1**

The second A1 is the point of the part: adding the same amount to every value slides the whole data set along without changing how spread out it is. Award it for "unchanged", "still 1.87", or " σ is not affected by adding a constant". Recomputing all eight new volumes and running the GDC again is a valid route to the same two marks.

(e) One standard deviation either side of the mean: $500 \pm 4.2720 \dots$, that is $495.7 < x < 504.3$.

Machine B volumes inside that interval: 499, 496, 503, 497, 499 — 5 bottles. **A1**

Accept the interval quoted as 495.73 to 504.27, or as $496 \leq x \leq 504$ if the working shows the correct boundaries. The excluded bottles are 494, 507 and 505. FT: award the A1 for a count made correctly from the candidate's own standard deviation in (b) — a candidate using $s_x = 4.57$ reaches 495.4 to 504.6 and the same 5 bottles.